Jorge Manuel Torre

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Education

Virginia Tech Blacksburg, VA
B.S. Computer Science, Overall GPA: 3.32/4.0, Major GPA: 3.50/4.0 May 2026
Relevant Coursework: Security of AI, Machine Learning (CS 4824), Linux Kernel Programming, Software Reverse Engineering, Computer Architecture, Computer Systems, Data Structures & Algorithms.

Experience

Systems Administrator / Infrastructure Engineer January 2019 – December 2023
Independent Systems Administration *Proxmox, Linux, NixOS, Docker, OPNsense*

- Administered and maintained a Linux infrastructure environment spanning Proxmox virtual machines, NixOS systems, and Docker-based services.
- Configured and managed network infrastructure using OPNsense, including firewall rules, port forwarding, DNS, routing, and service accessibility.
- Develop custom scripts, plugins, and server-side software to automate administrative tasks, service management, and system maintenance.
- Managed remote administration through SSH, access controls, service configuration, and secure network practices.
- Deployed and maintain containerized services using Docker, including networking, persistent storage, configuration, and service dependencies.
- Troubleshooted system, application, and network issues using logs, resource utilization, configuration analysis, and command-line tooling.
- Performed software upgrades, configuration changes, backups, and recovery procedures while maintaining service availability.

Projects

CHUD: Catastrophic Harmful Update Detection • GitHub January 2026 – April 2026
Python, PyTorch, LoRA, Llama-2-7b, Llama-Guard-3-8b, Unsloth *Team of 4 (CS Capstone)*

- Tested the LoX defense (Perin et al., COLM 2025) against an iterative LoRA fine-tuning attack on Llama-2-7b using benign GSM8k and adversarial BeaverTails samples.
- Built an Attack Success Rate (ASR) evaluation pipeline with Llama-Guard-3-8b as the judge model, calibrated it against AdvBench (99% accuracy, 2% FNR), and ran 30+ experiments.
- Found that LoX lowered ASR from 91–96% to 37–63% under direct harmful fine-tuning, but an iterative mixed-data attack raised ASR back to 89%.
- Set up x86-64 and aarch64 VT GPU cluster environments for the team and ran QLoRA fine-tuning evaluations for the final analysis.

Celeste-RL: Deep Reinforcement Learning for a Precision Platformer • GitHub • Website January 2026 – April 2026
Python, PyTorch, NumPy, matplotlib, CUDA

- Compared five RL methods (DQN, Behavioral Cloning, Hybrid, Curriculum, Dueling DQN with curiosity) trained to play Celeste Classic room 0 on the Pyleste emulator.
- Found that state representation mattered more than algorithm choice: replacing raw PICO-8 tile IDs with a 5-class encoding raised plain DQN completion from 8% to 57–68% with no other change.
- Built scripts for batched evaluation, comparison plots, side-by-side gameplay GIFs, a GitHub Pages results site, and a debugging devlog covering 9 iterations.

Experimental Home Lab January 2021 – Present
Proxmox, OPNsense, NixOS, Docker, CUDA

- Run a multi-VM Proxmox lab with NixOS / Docker workspaces and a CUDA-enabled training environment used for the ML projects above.

Leadership & Community

Team Lead, Database Management Systems (CS 4604) May 2025 – August 2025

- Led a 3-person team through schema design and SQL development, then integrated the team's work into the final submission.

Team Lead, Intermediate Software Design (CS 3704) January 2025 – April 2025

- Led a 5-person team through requirements gathering, sprint planning, code reviews, and final project delivery.

Garage32 (Community Project) March 2020 – December 2021

- Led development and distribution of a graphical asset pack for the Minecraft community, reaching **600+ downloads**; promotional content earned **30K+ likes** on a single TikTok post.

Skills

ML / AI: PyTorch, LoRA, Fine-Tuning, LLM Safety & Alignment, Reinforcement Learning (DQN, BC, Curriculum), Llama / Llama-Guard, Adversarial ML, Research Methodology
Languages: Python, C, C++, Java, SQL, JavaScript, Assembly (ARM, x86-64, RISC-V), Bash
Infrastructure & Tooling: CUDA, Docker, NixOS, Proxmox, Linux, Git, CI/CD
Also familiar with: React, Three.js, Django, PostgreSQL